TRUSTYHANDS

Service Marketplace Platform

Presented by:

Shruthi Deepika V - AP23110011584

Himakshi Ch - AP23110011596

Durga Sravanthi P - AP23110011597

Deekshitha Reddy L - AP23110011634

INTRODUCTION

TrustyHands is a web-based service marketplace designed to bridge the gap between customers and skilled workers. It provides a centralized platform where users can easily find trusted professionals for services such as home maintenance, repairs, and other daily needs.

In many real-world scenarios, finding reliable workers is time-consuming and uncertain. TrustyHands addresses this issue by offering a verified and structured environment where users can confidently book services and workers can showcase their skills.

PROBLEM STATEMENT

Core Challenges in Service Booking Systems

Challenge A: Difficulty in Finding Reliable Workers

Many users struggle to identify trustworthy service providers due to lack of proper verification systems. This leads to poor service quality and user dissatisfaction.

Challenge B: Lack of Centralized Platform

Existing solutions are scattered and unorganized, making it difficult for users to manage bookings and compare service providers efficiently.

Challenge C: Inefficient Booking Process

Traditional methods rely on manual communication, which leads to delays, confusion, and lack of proper tracking of service requests.

PROJECT OBJECTIVES

1. Reliable Service Platform Development
   Develop a structured and dependable platform that connects customers with verified workers efficiently.

2. Secure and Verified System
   Ensure that all workers are verified by administrators before being available for booking.

3. Efficient Booking and Communication
   Enable smooth booking processes and effective communication between users and workers.

SEPM OVERVIEW

Software Engineering Phases for TrustyHands

1. Requirements and Design
This phase involves gathering user requirements, defining system functionalities, and designing system architecture including database and user interfaces.

2. Implementation and Testing
Core modules such as authentication, booking system, and admin controls are developed. Testing ensures system reliability and performance.

3. Deployment and Maintenance
The system is deployed on cloud platforms and continuously updated based on user feedback and performance monitoring.

REQUIREMENTS GATHERING

Defining System Requirements

Functional Requirements

- User registration and login
- Service search and booking
- Worker profile management
- Admin verification system
- Feedback and review system

Non-Functional Requirements

- Secure authentication using JWT
- Fast response time
- User-friendly interface
- High system reliability

SYSTEM DESIGN

Architecture and Design Components

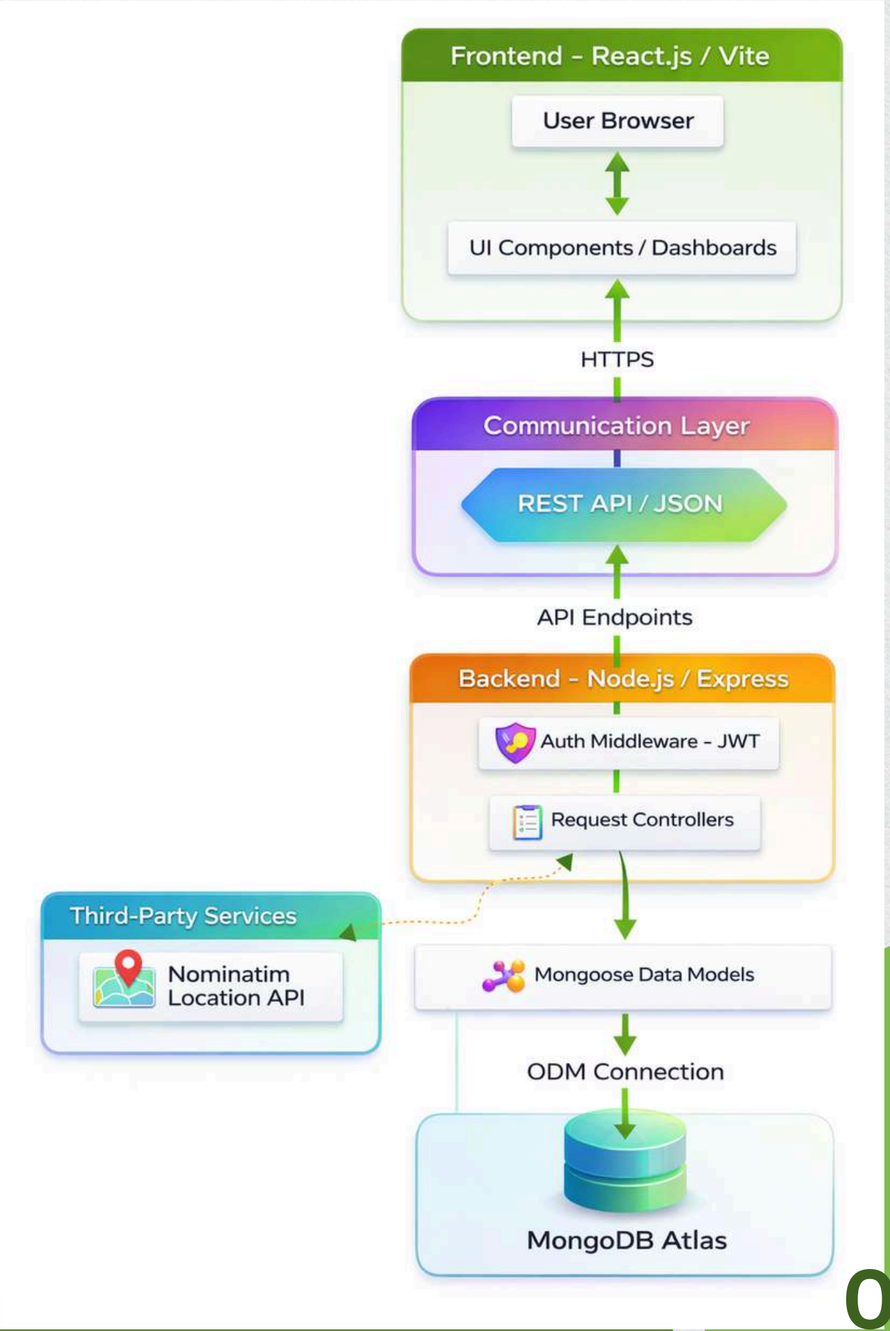
High-Level Architecture Components

Frontend: React-based user interface

Backend: Node.js and Express server

Database: MongoDB for data storage

API Layer: RESTful communication



IMPLEMENTATION AND TESTING

Development and Quality Assurance

Key Implementation Tasks

- Developing authentication system
- Building booking and service modules
- Creating admin dashboard
- Integrating database

Testing Approach

- Unit testing for individual modules
- Integration testing for system flow
- User acceptance testing

SYSTEM FEATURES

Core Functionalities of TrustyHands

- Secure login and registration
- Service search and filtering
- Booking and tracking system
- Reviews and ratings
- Messaging between users and workers
- Admin control panel

PROJECT CHALLENGES

Challenges Faced During Development

1. Challenge 1: Real-Time Data Handling
   Managing multiple user requests and booking updates efficiently.

2. Challenge 2: Security and Data Protection
   Ensuring secure handling of user data and authentication.

3. Challenge 3: System Scalability
   Designing a system that can handle increasing users and requests.

PROJECT TIMELINE

Planned Schedule and Milestones

WATERFALL MODEL WITH ITERATIVE FEEDBACK

1. Requirements Phase
   Duration: : 2 Weeks
   Requirement gathering and SRS preparation.

2. Design & Implementation
   Duration: : 6 Weeks
   System design and development of modules.

3. Testing & Deployment
   Duration: : 2 Weeks
   Testing, bug fixing, and final deployment.

RISK MANAGEMENT

Identifying and Mitigating Project Risks

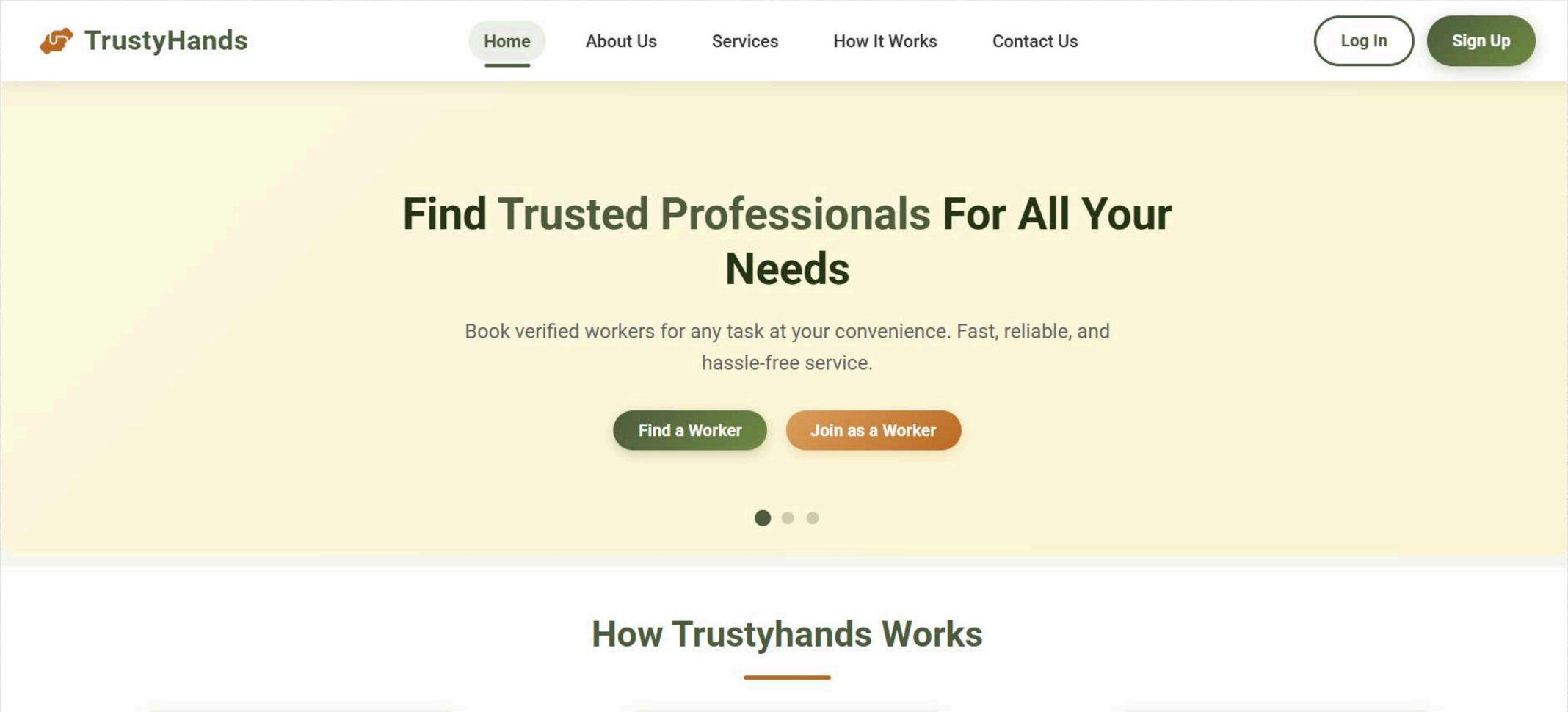
1. Risk 1: Technical Failure
   System crashes or server downtime may affect bookings.
   Mitigation: Use reliable servers, backups, and monitoring.

2. Risk 2: Data Security
   user data may be at risk of cyber attacks
   Mitigation: Use JWT authentication, encryption, and HTTPS.

3. Risk 3: Low User Adoption
   Users may not trust or understand the platform.
   Mitigation: Improve user trust through simple UI and verified workers.

RESULT

TrustyHands-website link



CONCLUSION

TrustyHands successfully provides a reliable and efficient platform for connecting customers with skilled workers. It simplifies service booking, improves trust through verification, and enhances user experience.

The system can be further improved with advanced features such as AI-based recommendations and mobile application support.

THANK YOU